

Step 2 **Configure OSPF and static routes.**

The loopback interface on R1 and connected interfaces on R1, R2, and R3 run in OSPF area 0. Routes of interfaces on R2 and R3 connecting to S1 are advertised to OSPF, but no OSPF neighbor relationships. The silent mode is therefore used.

To simulate PCs, R4 and R5 use default static routes pointing to 192.168.1.1 (VRRP virtual IP address).

Enable R1 to learn routes to 192.168.1.0, and enable R2 and R3 to learn routes to 1.1.1.1.

```
[R1]ospf 1
```

```
[R1-ospf-1]area 0
```

```
[R1-ospf-1-area-0.0.0.0]network 10.0.123.0 0.0.0.255
```

```
[R1-ospf-1-area-0.0.0.0]network 1.1.1.1 0.0.0.0
```

```
[R2]ospf 1
```

```
[R2-ospf-1]silent-interface GigabitEthernet 0/0/1
```

```
[R2-ospf-1]area 0
```

```
[R2-ospf-1-area-0.0.0.0]network 10.0.123.0 0.0.0.255
```

```
[R2-ospf-1-area-0.0.0.0]network 192.168.1.0 0.0.0.255
```

```
[R3]ospf 1
```

```
[R3-ospf-1]silent-interface GigabitEthernet 0/0/1
```

```
[R3-ospf-1]area 0
```

```
[R3-ospf-1-area-0.0.0.0]network 10.0.123.0 0.0.0.255
```

```
[R3-ospf-1-area-0.0.0.0]network 192.168.1.0 0.0.0.255
```

```
[R4]ip route-static 0.0.0.0 0.0.0.0 192.168.1.1
```

```
[R5]ip route-static 0.0.0.0 0.0.0.0 192.168.1.1
```

After the configuration is complete, check the routing table of R1, R2, and R4.

[R1]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 9 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.1.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.123.0/24	Direct	0	0	D	10.0.123.1	GigabitEthernet0/0/0
10.0.123.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.123.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	OSPF	10	2	D	10.0.123.3	GigabitEthernet0/0/0
	OSPF	10	2	D	10.0.123.2	GigabitEthernet0/0/0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R2]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 12 Routes : 12

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
------------------	-------	-----	------	-------	---------	-----------

1.1.1.1/32	OSPF	10	1	D	10.0.123.1	GigabitEthernet0/0/0
10.0.0.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.123.0/24	Direct	0	0	D	10.0.123.2	GigabitEthernet0/0/0
10.0.123.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.123.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.2	GigabitEthernet0/0/1
192.168.1.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
192.168.1.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R4]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 9 Routes : 9

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	60	0	RD	192.168.1.1	GigabitEthernet0/0/1
10.0.0.4/32	Direct	0	0	D	127.0.0.1	LoopBack0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.4	GigabitEthernet0/0/1
192.168.1.4/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
192.168.1.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1

```
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
```

The preceding output shows that R1 can learn routes to 192.168.1.0/24, R2 can learn routes to 1.1.1.1/32, and R4 has a default static route to 192.168.1.1.

Step 3 **Configure a VRRP group and the virtual IP address.**

Enable VRRP on interfaces of R2 and R3, and configure the VRID and virtual IP address.

```
[R2]interface GigabitEthernet 0/0/1
```

```
[R2-GigabitEthernet0/0/1]vrrp vrid 1 virtual-ip 192.168.1.1
```

R2 is configured first and becomes the master router if there is no other member in the VRRP group after a period of time.

```
[R3]interface GigabitEthernet 0/0/1
```

```
[R3-GigabitEthernet0/0/1]vrrp vrid 1 virtual-ip 192.168.1.1
```

After the configuration is complete, check the VRRP status on R2 and R3.

```
[R2]display vrrp
```

```
GigabitEthernet0/0/1 | Virtual Router 1
```

```
State : Master
```

```
Virtual IP : 192.168.1.1
```

```
Master IP : 192.168.1.2
```

```
PriorityRun : 100
```

```
PriorityConfig : 100
```

```
MasterPriority : 100
```

```
Preempt : YES Delay Time : 0 s
```

```
TimerRun : 1 s
```

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Create time : 2016-07-22 18:00:03

Last change time : 2016-07-22 18:00:07

[R3]display vrrp

GigabitEthernet0/0/1 | Virtual Router 1

State : Backup

Virtual IP : 192.168.1.1

Master IP : 192.168.1.2

PriorityRun : 100

PriorityConfig : 100

MasterPriority : 100

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Create time : 2016-07-22 18:03:16

Last change time : 2016-07-22 18:03:16

R2 is selected as the master router and R3 as the backup router. The priorities of master and slave routers are both 100. When R3 is started first, it becomes the master router, which is not expected.

Configure the VRRP priority and verify the active/standby switchover.

Configure VRRP priorities on R2 and R3. A greater priority value indicates a higher priority. Set VRRP priorities of R2 and R3 to 120 and 110, respectively.

```
[R2]interface GigabitEthernet 0/0/1
```

```
[R2-GigabitEthernet0/0/1]vrrp vrid 1 priority 120
```

```
[R3]interface GigabitEthernet 0/0/1
```

```
[R3-GigabitEthernet0/0/1]vrrp vrid 1 priority 110
```

Check the configuration after priorities are changed.

```
[R2]display vrrp
```

```
GigabitEthernet0/0/1 | Virtual Router 1
```

```
State : Master
```

```
Virtual IP : 192.168.1.1
```

```
Master IP : 192.168.1.2
```

```
PriorityRun : 120
```

```
PriorityConfig : 120
```

```
MasterPriority : 120
```

```
Preempt : YES    Delay Time : 0 s
```

```
TimerRun : 1 s
```

```
TimerConfig : 1 s
```

```
Auth type : NONE
```

```
Virtual MAC : 0000-5e00-0101
```

```
Check TTL : YES
```

Config type : normal-vrrp

Backup-forward : disabled

Create time : 2016-07-22 18:00:03

Last change time : 2016-07-22 18:00:07

[R3]display vrrp

GigabitEthernet0/0/1 | Virtual Router 1

State : Backup

Virtual IP : 192.168.1.1

Master IP : 192.168.1.2

PriorityRun : 110

PriorityConfig : 110

MasterPriority : 120

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Create time : 2016-07-22 18:03:16

Last change time : 2016-07-22 18:03:16

The preceding output shows that priorities of R2 and R3 have been changed successfully. By default, VRRP preemption is enabled. When the priority of R3 is changed to be higher, an active/standby switchover will be triggered.

Test the connectivity between R4 and R1.

[R4]ping 1.1.1.1

PING 1.1.1.1: 56 data bytes, press CTRL_C to break

Reply from 1.1.1.1: bytes=56 Sequence=1 ttl=254 time=57 ms

Reply from 1.1.1.1: bytes=56 Sequence=2 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=3 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=4 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 1.1.1.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/12/57 ms

The preceding output shows that the virtual gateway works properly and can forward data of the LAN where R4 is located to R1. Normally, the master router forwards data, so traffic passes through R2. To verify the switching status, perform the ping operation from R4 to R1 for a long time and shut down the interface of R2 connected to S1.

[R2]interface GigabitEthernet 0/0/1

[R2-GigabitEthernet0/0/1]shutdown

R4 discards two data packets during the switchover, and subsequent data is forwarded normally.

[R4]ping -c 1000 1.1.1.1

PING 1.1.1.1: 56 data bytes, press CTRL_C to break

Reply from 1.1.1.1: bytes=56 Sequence=1 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=2 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=3 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=4 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=5 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=6 ttl=254 time=1 ms

Request time out

Request time out

Reply from 1.1.1.1: bytes=56 Sequence=9 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=10 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=11 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=12 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=13 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=14 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=15 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=16 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=17 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=18 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=19 ttl=254 time=1 ms

Reply from 1.1.1.1: bytes=56 Sequence=20 ttl=254 time=1 ms

--- 1.1.1.1 ping statistics ---

20 packet(s) transmitted

18 packet(s) received

10.00% packet loss

round-trip min/avg/max = 1/1/1 ms

R3 becomes the master router after the switchover.

[R3]display vrrp

GigabitEthernet0/0/1 | Virtual Router 1

State : Master

Virtual IP : 192.168.1.1

Master IP : 192.168.1.3

PriorityRun : 110

PriorityConfig : 110

MasterPriority : 110

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Create time : 2016-07-22 18:03:16

Last change time : 2016-07-22 18:29:41

Configure VRRP to monitor the uplink.

The VRRP active/standby switchover is implemented by listening to Advertisement packets. If the backup router cannot listen to messages of the master router or has a higher priority, the backup router preempts to be the master router (no preemption delay by default).

If the fault occurs on the uplink link, the active/standby switchover is not performed. In this case, all Internet access traffic cannot be forwarded after reaching R2. VRRP is enabled to monitor the uplink. When the uplink fails, R2 automatically reduces its priority. R3 preempts to be the master router, and traffic is switched to the backup router and backup uplink.

Before configuring VRRP to monitor the uplink, restore the link that is shut down.

Configure VRRP to monitor the uplink interface and set the value by which the

priority decreases to 30. That is, when the link fails, the priority of R2 becomes 90, which is lower than the priority of R3 (110).

```
[R2]interface GigabitEthernet 0/0/1
```

```
[R2-GigabitEthernet0/0/1]undo shutdown
```

```
[R2-GigabitEthernet0/0/1]vrrp vrid 1 track interface GigabitEthernet 0/0/0 reduced 30
```

Check the configuration.

```
[R2]display vrrp
```

```
GigabitEthernet0/0/1 | Virtual Router 1
```

```
State : Master
```

```
Virtual IP : 192.168.1.1
```

```
Master IP : 192.168.1.2
```

```
PriorityRun : 120
```

```
PriorityConfig : 120
```

```
MasterPriority : 120
```

```
Preempt : YES    Delay Time : 0 s
```

```
TimerRun : 1 s
```

```
TimerConfig : 1 s
```

```
Auth type : NONE
```

```
Virtual MAC : 0000-5e00-0101
```

```
Check TTL : YES
```

```
Config type : normal-vrrp
```

```
Backup-forward : disabled
```

```
Track IF : GigabitEthernet0/0/0    Priority reduced : 30
```

```
IF state : UP
```

```
Create time : 2016-07-25 17:14:56 UTC-08:00
```

```
Last change time : 2016-07-25 17:32:27 UTC-08:00
```

Perform the ping operation on R4 for a long time and shut down the uplink interface on R2.

```
[R2]interface GigabitEthernet 0/0/0
```

```
[R2-GigabitEthernet0/0/0]shutdown
```

```
[R2]display vrrp
```

GigabitEthernet0/0/1 | Virtual Router 1

State : Backup

Virtual IP : 192.168.1.1

Master IP : 192.168.1.3

PriorityRun : 90

PriorityConfig : 120

MasterPriority : 110

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Track IF : GigabitEthernet0/0/0 Priority reduced : 30

IF state : DOWN

Create time : 2016-07-25 17:14:56 UTC-08:00

Last change time : 2016-07-25 19:57:46 UTC-08:00

```
[R3]display vrrp
```

GigabitEthernet0/0/1 | Virtual Router 1

State : Master

Virtual IP : 192.168.1.1
Master IP : 192.168.1.3
PriorityRun : 110
PriorityConfig : 110
MasterPriority : 110
Preempt : YES Delay Time : 0 s
TimerRun : 1 s
TimerConfig : 1 s
Auth type : NONE
Virtual MAC : 0000-5e00-0101
Check TTL : YES
Config type : normal-vrrp
Backup-forward : disabled
Create time : 2016-07-25 17:20:00 UTC-08:00
Last change time : 2016-07-25 19:56:24 UTC-08:00

R3 becomes the master router and traffic is successfully switched to R3.

Restore the uplink and priority of R2. R2 preempts to be the master router again. During preemption, few packets are discarded on R4. This is because OSPF routes are not converged rapidly. For details about route convergence acceleration, see the OSPF experiment.

```
[R2]interface GigabitEthernet 0/0/0
```

```
[R2-GigabitEthernet0/0/0]undo shutdown
```

```
[R2]display vrrp
```

```
GigabitEthernet0/0/1 | Virtual Router 1
```

```
State : Master
```

```
Virtual IP : 192.168.1.1
```

Master IP : 192.168.1.2

PriorityRun : 120

PriorityConfig : 120

MasterPriority : 120

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Track IF : GigabitEthernet0/0/0 Priority reduced : 30

IF state : UP

Create time : 2016-07-25 17:14:56 UTC-08:00

Last change time : 2016-07-25 20:04:40 UTC-08:00

When the interface goes Up, the OSPF neighbor relationship needs to be reestablished on the uplink interface of R2. If OSPF fast convergence is not configured, data cannot be forwarded for several seconds. You are advised to set the preemption delay to be longer than the OSPF convergence time during the switchback.

```
[R2]interface GigabitEthernet 0/0/1
```

```
[R2-GigabitEthernet0/0/1]vrrp vrid 1 preempt-mode timer delay 10
```

Check the VRRP configurations again. You can see that the preemption delay has been configured successfully.

```
[R2]display vrrp
```

GigabitEthernet0/0/1 | Virtual Router 1

State : Master

Virtual IP : 192.168.1.1

Master IP : 192.168.1.2

PriorityRun : 120

PriorityConfig : 120

MasterPriority : 120

Preempt : YES Delay Time : 10 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Track IF : GigabitEthernet0/0/0 Priority reduced : 30

IF state : UP

Create time : 2016-07-25 17:14:56 UTC-08:00

Last change time : 2016-07-25 20:04:40 UTC-08:00

Step 4 **Configure load balancing of multiple VRRP groups.**

Normally, the master device forwards all traffic, and the backup device is idle.

To implement dual-gateway load balancing, configure multiple VRRP groups.

Configure VRRP group 1 on R2 and R3, set the virtual IP address to 192.168.1.1, and configure R2 as the master device. Configure VRRP group 2, set the virtual address to 192.168.1.254, and configure R3 as the master device. Configure the default gateway address pointing to 192.168.1.1 for R4 and the default gateway address pointing to 192.168.1.254 for R5. Internet access traffic can be load balanced to two

gateways.

The configuration is as follows.

```
[R2]interface GigabitEthernet 0/0/1
```

```
[R2-GigabitEthernet0/0/1]vrrp vrid 2 virtual-ip 192.168.1.254
```

```
[R2-GigabitEthernet0/0/1]vrrp vrid 2 priority 110
```

```
[R3]interface GigabitEthernet 0/0/1
```

```
[R3-GigabitEthernet0/0/1]vrrp vrid 2 virtual-ip 192.168.1.254
```

```
[R3-GigabitEthernet0/0/1]vrrp vrid 2 priority 120
```

```
[R3-GigabitEthernet0/0/1]vrrp vrid 2 track interface GigabitEthernet0/0/0 reduced 30
```

```
[R5]undo ip route-static 0.0.0.0 0.0.0.0 192.168.1.1
```

```
[R5]ip route-static 0.0.0.0 0.0.0.0 192.168.1.254
```

Check load balancing of two VRRP groups on R2 and R3.

```
<R2>display vrrp
```

```
GigabitEthernet0/0/1 | Virtual Router 1
```

```
State : Master
```

```
Virtual IP : 192.168.1.1
```

```
Master IP : 192.168.1.2
```

```
PriorityRun : 120
```

```
PriorityConfig : 120
```

```
MasterPriority : 120
```

```
Preempt : YES    Delay Time : 10 s
```

```
TimerRun : 1 s
```

```
TimerConfig : 1 s
```

```
Auth type : NONE
```

```
Virtual MAC : 0000-5e00-0101
```


Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Track IF : GigabitEthernet0/0/0 Priority reduced : 30

IF state : UP

Create time : 2016-07-25 17:14:56 UTC-08:00

Last change time : 2016-07-25 20:04:40 UTC-08:00

GigabitEthernet0/0/1 | Virtual Router 2

State : Backup

Virtual IP : 192.168.1.254

Master IP : 192.168.1.3

PriorityRun : 110

PriorityConfig : 110

MasterPriority : 120

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0102

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Create time : 2016-07-25 17:15:54 UTC-08:00

Last change time : 2016-07-25 17:20:30 UTC-08:00

<R3>display vrrp

GigabitEthernet0/0/1 | Virtual Router 1

State : Backup

Virtual IP : 192.168.1.1
 Master IP : 192.168.1.2
 PriorityRun : 110
 PriorityConfig : 110
 MasterPriority : 120
 Preempt : YES Delay Time : 0 s
 TimerRun : 1 s
 TimerConfig : 1 s
 Auth type : NONE
 Virtual MAC : 0000-5e00-0101
 Check TTL : YES
 Config type : normal-vrrp
 Backup-forward : disabled
 Create time : 2016-07-25 17:20:00 UTC-08:00
 Last change time : 2016-07-25 20:03:15 UTC-08:00

GigabitEthernet0/0/1 | Virtual Router 2

State : Master
 Virtual IP : 192.168.1.254
 Master IP : 192.168.1.3
 PriorityRun : 120
 PriorityConfig : 120
 MasterPriority : 120
 Preempt : YES Delay Time : 0 s
 TimerRun : 1 s
 TimerConfig : 1 s
 Auth type : NONE
 Virtual MAC : 0000-5e00-0102
 Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Track IF : GigabitEthernet0/0/0 Priority reduced : 30

IF state : UP

Create time : 2016-07-25 17:20:14 UTC-08:00

Last change time : 2016-07-25 17:20:23 UTC-08:00

Perform the tracert operation to check the gateways that process data destined for the two default routes. You can see that data sent by R4 is forwarded by the master device in VRRP group 1 and data sent by R5 is forwarded by the master device in VRRP group 2.

Enable R1 to send ICMP Port Unreachable packets.

[R1]icmp port-unreachable send

<R4>tracert 1.1.1.1

tracert to 1.1.1.1(1.1.1.1), max hops: 30 ,packet length: 40,press CTRL_C t
o break

1 192.168.1.2 80 ms 40 ms 40 ms

2 10.0.123.1 100 ms 70 ms 70 ms

<R5>tracert 1.1.1.1

tracert to 1.1.1.1(1.1.1.1), max hops: 30 ,packet length: 40,press CTRL_C t
o break

```
1 192.168.1.3 50 ms 30 ms 50 ms
```

```
2 10.0.123.1 60 ms 90 ms 60 ms
```

Check the switchover when the uplink fails.

```
[R2]interface GigabitEthernet 0/0/0
```

```
[R2-GigabitEthernet0/0/0]shutdown
```

```
<R4>tracert 1.1.1.1
```

```
tracert to 1.1.1.1(1.1.1.1), max hops: 30 ,packet length: 40,press CTRL_C t
o break
```

```
1 192.168.1.3 50 ms 40 ms 50 ms
```

```
2 10.0.123.1 70 ms 80 ms 50 ms
```

```
<R5>tracert 1.1.1.1
```

```
tracert to 1.1.1.1(1.1.1.1), max hops: 30 ,packet length: 40,press CTRL_C t
o break
```

```
1 192.168.1.3 40 ms 50 ms 40 ms
```

```
2 10.0.123.1 70 ms 100 ms 90 ms
```

Check the status of the two VRRP groups.

```
<R2>display vrrp
```

GigabitEthernet0/0/1 | Virtual Router 1

State : Backup

Virtual IP : 192.168.1.1

Master IP : 192.168.1.3

PriorityRun : 90

PriorityConfig : 120

MasterPriority : 110

Preempt : YES Delay Time : 10 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Track IF : GigabitEthernet0/0/0 Priority reduced : 30

IF state : DOWN

Create time : 2016-07-25 17:14:56 UTC-08:00

Last change time : 2016-07-25 20:48:28 UTC-08:00

GigabitEthernet0/0/1 | Virtual Router 2

State : Backup

Virtual IP : 192.168.1.254

Master IP : 192.168.1.3

PriorityRun : 110

PriorityConfig : 110

MasterPriority : 120

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0102

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Create time : 2016-07-25 17:15:54 UTC-08:00

Last change time : 2016-07-25 17:20:30 UTC-08:00

<R3>display vrrp

GigabitEthernet0/0/1 | Virtual Router 1

State : Master

Virtual IP : 192.168.1.1

Master IP : 192.168.1.3

PriorityRun : 110

PriorityConfig : 110

MasterPriority : 110

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0101

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Create time : 2016-07-25 17:20:00 UTC-08:00

Last change time : 2016-07-25 20:46:42 UTC-08:00

GigabitEthernet0/0/1 | Virtual Router 2

State : Master

Virtual IP : 192.168.1.254

Master IP : 192.168.1.3

PriorityRun : 120

PriorityConfig : 120

MasterPriority : 120

Preempt : YES Delay Time : 0 s

TimerRun : 1 s

TimerConfig : 1 s

Auth type : NONE

Virtual MAC : 0000-5e00-0102

Check TTL : YES

Config type : normal-vrrp

Backup-forward : disabled

Track IF : GigabitEthernet0/0/0 Priority reduced : 30

IF state : UP

Create time : 2016-07-25 17:20:14 UTC-08:00

Last change time : 2016-07-25 17:20:23 UTC-08:00

Normally, R2 and R3 load balance traffic. When R2 is faulty, R3 takes over all traffic on R2. In this case, load balancing of two VRRP groups is configured.

Device Configuration

<R1>display current-configuration

#

sysname R1

#

```
interface GigabitEthernet0/0/0
  ip address 10.0.123.1 255.255.255.0
#
interface LoopBack0
  ip address 1.1.1.1 255.255.255.255
#
ospf 1
  area 0.0.0.0
    network 1.1.1.1 0.0.0.0
    network 10.0.123.0 0.0.0.255
#
return
```

<R2> display current-configuration

```
#
sysname R2
#
interface GigabitEthernet0/0/0
  shutdown
  ip address 10.0.123.2 255.255.255.0
#
interface GigabitEthernet0/0/1
  ip address 192.168.1.2 255.255.255.0
  vrrp vrid 1 virtual-ip 192.168.1.1
  vrrp vrid 1 priority 120
  vrrp vrid 1 preempt-mode timer delay 10
  vrrp vrid 1 track interface GigabitEthernet0/0/0 reduced 30
  vrrp vrid 2 virtual-ip 192.168.1.254
```



```
vrrp vrid 2 priority 110

#

ospf 1

silent-interface GigabitEthernet0/0/1

area 0.0.0.0

network 10.0.123.0 0.0.0.255

network 192.168.1.0 0.0.0.255

#

return
```

<R3>display current-configuration

```
#

sysname R3

#

interface GigabitEthernet0/0/0

ip address 10.0.123.3 255.255.255.0

#

interface GigabitEthernet0/0/1

ip address 192.168.1.3 255.255.255.0

vrrp vrid 1 virtual-ip 192.168.1.1

vrrp vrid 1 priority 110

vrrp vrid 2 virtual-ip 192.168.1.254

vrrp vrid 2 priority 120

vrrp vrid 2 track interface GigabitEthernet0/0/0 reduced 30

#

ospf 1

silent-interface GigabitEthernet0/0/1

area 0.0.0.0
```

```
network 10.0.123.0 0.0.0.255
network 192.168.1.0 0.0.0.255
#
return
```

<R4> display current-configuration

```
#
sysname R4
#
interface GigabitEthernet0/0/1
ip address 192.168.1.4 255.255.255.0
#
ip route-static 0.0.0.0 0.0.0.0 192.168.1.1
#
return
```

<R5> display current-configuration

```
#
sysname R5
#
interface GigabitEthernet0/0/1
ip address 192.168.1.5 255.255.255.0
#
ip route-static 0.0.0.0 0.0.0.0 192.168.1.254
#
return
```

Chapter 6 BFD Configuration

Lab 6-1 Association Between BFD and Static Routes

Learning Objectives

The objectives of this lab are to learn and understand:

- How to associate Bidirectional Forwarding Detection (BFD) with static routes to implement floating routes

Topology

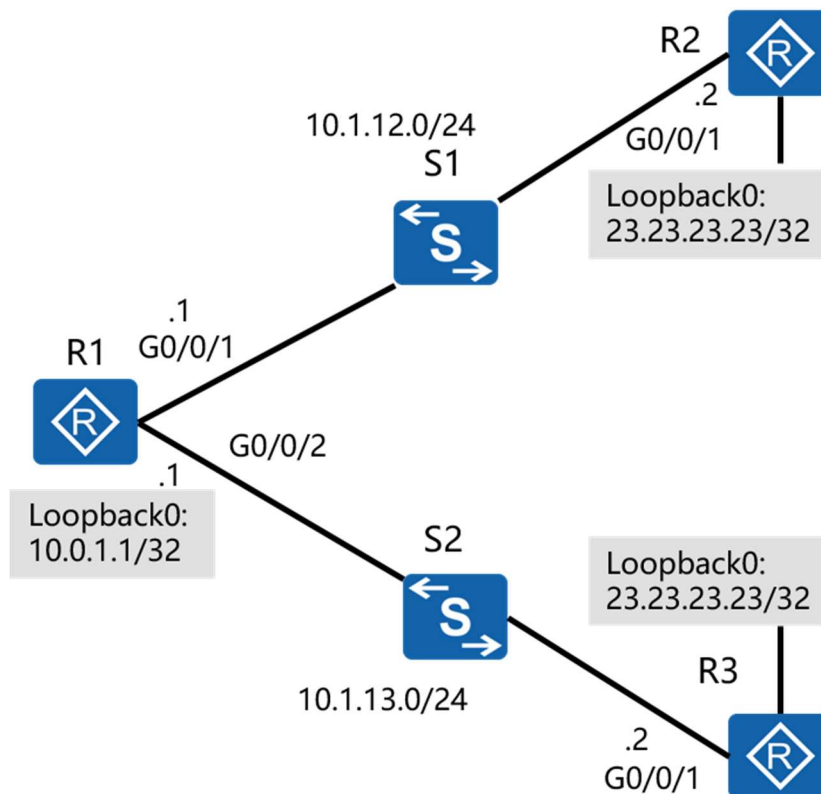


Figure 6-1 Networking of association between BFD and static routes

Scenario

R1 is connected to R2 and R3 through S1 and S2. Devices are connected through static routes, and packets can reach the target network 23.23.23.23/32 through R2 or